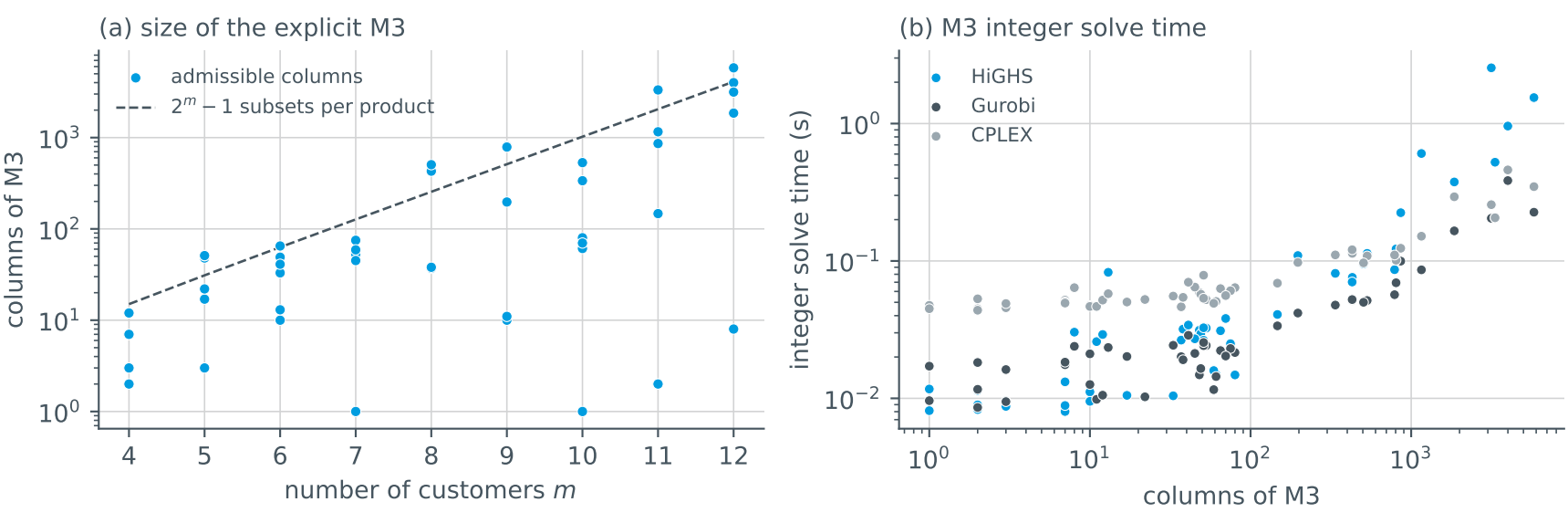
(a) size of the explicit M3
admissible columns
$2^m - 1$ subsets per product
columns of M3
10^0
10^1
10^2
10^3
4
5
6
7
8
9
10
11
12
number of customers m
(b) M3 integer solve time
HiGHS
Gurobi
CPLEX
integer solve time (s)
10^0
10^{-1}
10^{-2}
10^0
10^1
10^2
10^3
columns of M3